

## Sparse Graphical Models for Binary Variables

### Description

We consider the problem of estimating an undirected graphical model for multivariate binary data. More precisely, we aim to estimate the probability mass function  $p(x)$  of a binary random vector  $X = (X_1, \dots, X_n)^\top \in \{-1, 1\}^n$  based on  $m$  training samples  $\hat{x}^k \in \{-1, 1\}^n$ ,  $k = 1, \dots, m$ , drawn independently from  $p(x)$ .

As an example, we might want to learn semantic connections between different words indexed by  $i = 1, \dots, n$ . In this context, the random variable  $X_i$  indicates whether word  $i$  appears in a random document (*i.e.*,  $X_i = 1$  if word  $i$  appears and  $X_i = -1$  otherwise) and the training samples represent  $m$  randomly chosen documents (*i.e.*,  $\hat{x}_i^k = 1$  if word  $i$  appears in document  $k$  and  $\hat{x}_i^k = -1$  otherwise).

An Ising model for the probability mass function  $p(x)$  is given by

$$p_\theta(x) = \exp \left( \sum_{i=1}^n \theta_{ii} x_i + \sum_{\substack{i,j=1 \\ j \neq i}}^n x_i \theta_{ij} x_j - A(\theta) \right),$$

where  $\theta \in \mathbb{S}^n$  and

$$A(\theta) = \log \left( \sum_{x \in \{-1, 1\}^n} \exp \left( \sum_{i=1}^n \theta_{ii} x_i + \sum_{\substack{i,j=1 \\ j \neq i}}^n x_i \theta_{ij} x_j \right) \right)$$

is a normalization constant. The parameters  $\theta_{ij} = \theta_{ji}$ ,  $i \neq j$ , capture the dependence between the random variables  $X_i$  and  $X_j$ . In fact, one can show that if  $\theta_{ij} = 0$ , then  $X_i$  and  $X_j$  are conditionally independent given all other random variables  $X_k$ ,  $k \in \{1, \dots, n\} \setminus \{i, j\}$ .

The Ising model can conveniently be represented as a conditional dependence graph, where the nodes correspond to the different random variables  $X_i$ ,  $i = 1, \dots, n$ . We connect two random variables  $X_i$  and  $X_j$  with an edge whenever  $\theta_{ij} \neq 0$ , and  $\theta_{ij}$  can be viewed as the weight of the edge.

The likelihood to observe the training samples  $\hat{x}^k$ ,  $k = 1, \dots, m$ , is given by  $l(\theta) = \prod_{k=1}^m p_\theta(\hat{x}^k)$ , while the log-likelihood function can be represented as

$$L(\theta) = \log(l(\theta)) = \sum_{k=1}^m \log(p_\theta(\hat{x}^k)) = \sum_{k=1}^m \left( \sum_{i=1}^n \theta_{ii} \hat{x}_i^k + \sum_{\substack{i,j=1 \\ j \neq i}}^n \hat{x}_i^k \theta_{ij} \hat{x}_j^k \right) - mA(\theta).$$

In this exercise, we are interested in estimating a sparse graphical model, that is, we wish to identify a matrix  $\theta \in \mathbb{S}^n$  that has a high empirical (log-) likelihood and only few non-zero off-diagonal entries, *i.e.*, the corresponding conditional dependence graph should have few edges. To this end, we will minimize the sum of the negative empirical log-likelihood and a sparsity-inducing  $\ell_1$ -norm penalty term  $\rho \|\theta - \text{diag}(\theta_{11}, \dots, \theta_{nn})\|_1$ , where  $\rho \geq 0$  is the penalty weight,  $\|Q\|_1 := \sum_{i=1}^n \sum_{j=1}^n |Q_{ij}|$  for an arbitrary matrix  $Q \in \mathbb{S}^n$ , and  $\text{diag}(\theta_{11}, \dots, \theta_{nn})$  represents the diagonal matrix with diagonal entries  $\theta_{11}, \dots, \theta_{nn}$ . Thus, we aim to solve the convex optimization problem

$$\underset{\theta \in \mathbb{S}^n}{\text{minimize}} \quad A(\theta) - \left( \sum_{i=1}^n \theta_{ii} \hat{\mu}_i + \sum_{\substack{i,j=1 \\ j \neq i}}^n \theta_{ij} \hat{s}_{ij} \right) + \rho \|\theta - \text{diag}(\theta_{11}, \dots, \theta_{nn})\|_1, \quad (1)$$

where we denote by  $\hat{\mu} = \frac{1}{m} \sum_{k=1}^m \hat{x}^k$  the sample mean and by  $\hat{s} = \frac{1}{m} \sum_{k=1}^m \hat{x}^k (\hat{x}^k)^\top$  the sample second-order moment matrix.

Even though problem (1) is convex, it is hard to solve because the normalization constant  $A(\theta)$  involves a sum of  $2^n$  terms. In [1, Lemma 5] it has been shown, however, that  $A(\theta)$  admits a more tractable convex upper bound of the form

$$\bar{A}(\theta) = \frac{n}{2} \log\left(\frac{e\pi}{2}\right) - \frac{1}{2}(n+1) - \frac{1}{2} \left( \max_{v \in \mathbb{R}^{n+1}} v^\top r + \log \det \left( - (R(\theta) + \text{diag}(v)) \right) \right),$$

where  $e$  denotes the base of the natural logarithm,  $r = (1, \frac{4}{3}, \dots, \frac{4}{3}) \in \mathbb{R}^{n+1}$  and

$$R(\theta) = \begin{bmatrix} 0 & \theta_{11} & \theta_{22} & \dots & \theta_{nn} \\ \theta_{11} & 0 & 2\theta_{12} & \dots & 2\theta_{1n} \\ \theta_{22} & 2\theta_{21} & 0 & \dots & 2\theta_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \theta_{nn} & 2\theta_{n1} & 2\theta_{n2} & \dots & 0 \end{bmatrix}.$$

This result motivates us to solve the approximate estimation problem

$$\underset{\theta \in \mathbb{S}^n}{\text{minimize}} \quad \bar{A}(\theta) - \left( \sum_{i=1}^n \theta_{ii} \hat{\mu}_i + \sum_{\substack{i,j=1 \\ j \neq i}}^n \theta_{ij} \hat{s}_{ij} \right) + \rho \|\theta - \text{diag}(\theta_{11}, \dots, \theta_{nn})\|_1. \quad (2)$$

## Questions

1. Verify that problems (1) and (2) are convex.
2. The data files `docs.npy` and `words.txt` available from Moodle contains 20,000 newsgroup documents, partitioned (nearly) evenly across 20 different newsgroups. Each newsgroup document consists of 60 binary entries that correspond to 60 distinct words. Solve problem (2) for this data set with  $\rho = 10^{-3}, 5 \times 10^{-3}, 10^{-2}$  using the solver MOSEK. A skeleton of the code you will have to implement is provided in the Python file `p1q2.py`.

## Bonus Question (not related to optimization and therefore not examinable)

Prove that if  $\theta_{ij} = \theta_{ji} = 0$ , the random variables  $X_i$  and  $X_j$  are indeed conditionally independent given  $X_k$ ,  $k \in \{1, \dots, n\} \setminus \{i, j\}$ . Specifically, show that if  $\theta_{ij} = \theta_{ji} = 0$ , then

$$p_\theta(x_i, x_j | x_{-(i,j)}) = p_\theta(x_i | x_{-(i,j)}) p_\theta(x_j | x_{-(i,j)}),$$

where  $x_{-(i,j)}$  denotes the vector  $x$  without its  $i$ -th and  $j$ -th components.

## References

- [1] Onureena Banerjee, Laurent El Ghaoui, and Alexandre d'Aspremont. Model selection through sparse maximum likelihood estimation for multivariate Gaussian or binary data. *Journal of Machine Learning Research*, 9:485–516, 2008.